

Introduction and the Machine Learning Workflow — Key Concepts

Lesson 1 — Technologies for Artificial Intelligence

Fabio Antonini — Università degli Studi dell'Aquila

25 September 2026 · a one-page index

Contents

What learning is	1
Why it can go wrong	1
When not to reach for it	2
The three kinds of learning	2
The workflow, and its order	3
Four ways a result misleads	3
Limits worth stating on day one	3

An index, not a summary. Each entry says what a term means and where the handout treats it properly. Use this to find your way back into the text, or to check before an exercise that there is no word here you could not define.

What learning is

Learning from data. Supplying examples and letting a procedure search for a rule consistent with them, instead of stating the rule yourself. The method of choice when you can recognise the answer but cannot articulate it. → § 2

Input and output space. The set of things you observe and the set of answers you want, written $\mathcal{X}$ and $\mathcal{Y}$. → § 2.1

Loss function. $L(\hat{y}, y)$: the cost of answering $\hat{y}$ when the truth is y . Where you state what counts as a bad mistake, before any model exists. → § 2.1

Expected risk. The average loss over all data the world might produce, including data that does not exist yet. The quantity we want small, and the one we cannot compute. → § 2.1

Empirical risk. The same average taken over the finite sample you hold. What you can compute, and what fitting actually minimises. → § 2.1

Empirical risk minimisation. Choosing the model that makes the empirical risk small. Nearly every method in this course is an instance of it. → § 2.1

Why it can go wrong

Overfitting. Driving the empirical risk to zero by memorising the sample rather than learning the pattern. The central difficulty of the field. → § 2.2

Generalisation. How a model performs on data it has never seen — the only question that matters, and the one a training score cannot answer. → § 2.2

Train/test split. Holding data out, never letting the fitting procedure touch it, and measuring there. The entire reason a test set exists. → § 2.2

Unbiased estimation. A test set independent of the fitted model gives a test score that is right *on average*. A statement about the procedure, not a guarantee about your one number. → § 2.3

Independence of the test set. The condition the whole argument rests on. It breaks the moment a test row influences any decision — a mean, a ranking, a comparison. → § 2.3

Standard error of a proportion. Roughly $\sqrt{p(1-p)/m}$: how much a measured accuracy would move if you drew a different test set of the same size. → § 2.4

Why it has that form. One test row is a yes/no draw of variance $p(1-p)$, and averaging m independent draws divides a variance by m . Lesson 5 asks what happens when they are not independent. → § 2.4

Validation set. The third split, used for choosing between models and settings, so that the test set stays untouched until the end. → § 2.5

When not to reach for it

Known rules. If you can state the rule, write it. A classifier for something you can specify is a worse program than the specification. → § 3

Unrepresentative data. A model learns the distribution it was trained on; measured performance says little if deployment draws from another one. → § 3

Unrecoverable, unreviewed errors. Errors are certain; whether they can be undone, and whether a human sees the output first, are properties of the application and the deployment, decidable before any model exists. → § 3

The three kinds of learning

Supervised learning. Each example carries a target recorded because somebody wanted it — by a person or an instrument. The answer is hidden only to evaluate, on test rows that simulate the cases to come. Classification when the target is a category, regression when it is continuous. → § 5.1

Unsupervised learning. No targets: the goal is structure — groups, a lower-dimensional description, unusual points. There is still a loss; what is missing is something to check the answer against. → § 5.2

Self-supervised learning. The target is manufactured from the input by hiding part of it — hiding to train rather than to evaluate — and nobody wants the prediction for itself; the point is what transfers. Mechanically identical to a supervised regression, so the two are told apart by why the data is hidden, not by the code. → § 5.3

Pretext task. A target invented to force a representation: solving it is not the goal, and its score is not a measure of anything wanted. → § 5.3

The workflow, and its order

Framing. Deciding what is predicted, from what, and which mistake is worse — before modelling, because a metric chosen after seeing results is chosen to flatter them. → § 6

Baseline. The simplest thing that could work: the majority class, the mean, a threshold on one variable. Without one, a score has no meaning. → § 6

Pipeline. An object that binds preprocessing to the model, so that every step which learns from data is fitted on training data alone — structurally, not by remembering to. → § 6

Error analysis. Asking where a model fails and whether the failure is systematic. Worth more than another decimal place of accuracy. → § 6

Four ways a result misleads

Data leakage. Information from the test set reaching the training procedure — through a scaler, a feature selection, or a column recorded after the outcome. No error is raised. → § 7

Class imbalance. When the interesting class is rare, accuracy measures the class you are not interested in. → § 7

Shortcut feature. A column that is a consequence of the label rather than a predictor of it, and which will not exist when a prediction is needed. No metric detects it; knowing how the data was recorded does. → § 7

Single-split noise. One split is one draw from a distribution. A result quoted with no indication of variability is incomplete. → § 7

Limits worth stating on day one

Inherited bias. A model is a compressed summary of its training data, injustices included, and it reproduces them with an appearance of objectivity that makes them harder to contest. → § 8

Correlation, not cause. Nothing in empirical risk minimisation separates a cause from a coincidence that predicts well in this sample. → § 8

Beyond accuracy. Explainability, recoverability of errors, and who bears their cost are engineering requirements that belong in the framing step. → § 8